Paper id: 251019

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VI) THEORY EXAMINATION 2024-25
DATA COMPRESSION

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **02 x 7 = 14**

Q no.	Question	CO	Level
a.	What is data compression? Explain with the help of block diagram.	1	K2
b.	What are some real-world applications of data compression?	1	K2
c.	What is the principle behind Huffman coding?	1	K2
d.	What is the role of the Linde-Buzo-Gray (LBG) algorithm in vector quantization?	5	K2
e.	Differentiate between lossless and lossy compression with examples.	1	K2
f.	List the distortion criteria commonly used in quantization and explain their importance	5	K2
g.	Determine the Golomb code representations of 9 and 13 when m=5.	1	K2

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	A networking company uses a compression technique to encode the message before transmitting over the network. Suppose the message contains the following characters with their frequency: **Character** / **Frequency** **a** 5 **b** 9 **c** 12 **d** 13 **e** 16 **f** 45 Note that each character in input message takes 1 byte. If the compression technique used is Huffman Coding, how many bits will be saved in the message?	2	K2
b.	What is a Golomb code? How is it different from Rice code?	1	K2
c.	Describe the function of the Burrows-Wheeler Transform in compression.	4	K2
d.	What are the key elements of the quantization problem?	5	K2
e.	What is the role of the Linde-Buzo-Gray (LBG) algorithm in vector quantization?	5	K2

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Differentiate between LZ77 and LZ78 dictionary compression approaches.	3	K2
b.	Write a short note on: i. Compression Standards ii. Prediction by Partial Matching (PPM) iii. Burrows-Wheeler Transform (BWT) iv. Audio Compression v. Markov Model	1	K2

QP25EP1_029 | 13-Jun-2025 1:47:33 PM | 103.77.186.12

Paper id: 251019

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VI) THEORY EXAMINATION 2024-25
DATA COMPRESSION

TIME: 3 HRS **M.MARKS: 70**

4. Attempt any *one* part of the following: 07 x 1 = 07

a.	Encode the string **"ABABABA"** using LZW compression. Assume the initial dictionary contains all single uppercase letters **(A–Z)** with ASCII values **(A=65, B=66, ...)**. Provide the output code sequence.	1	K2
b.	Write a short note on: i. Entropy ii. Compression Ratio iii. Prefix Code iv. Run-Length Encoding (RLE) v. Signal-to-Noise Ratio (SNR)	1	K2

5. Attempt any *one* part of the following: 07 x 1 = 07

a.	Compare Binary coding with Huffman coding and explain their differences in performance.	2	K2
b.	Explain the LZ77 and LZ78 approaches and how they are used in UNIX file compression.	3	K2

6. Attempt any *one* part of the following: 07 x 1 = 07

a.	Explain the process of scalar quantization and discuss the models used for it.	4	K2
b.	Describe the types of quantizers: uniform, adaptive, and non-uniform with examples.	4	K2

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	Describe the differences between structured and tree-structured vector quantizers.	5	K2
b.	Explain the working of the Linde-Buzo-Gray algorithm for vector quantization.	4	K2

QP25EP1_029 | 13-Jun-2025 1:47:33 PM | 103.77.186.12

QP25EP1_029 | 13-Jun-2025 1:47:33 PM | 103.77.186.12